

A counterexample to the claimed closed form for grid burning numbers

Rule-3 disproof submission for conjecture 00000003949

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1 The conjecture and the burning number

Definition 1 (Graph burning). The *burning number* $b(G)$ of a finite graph G is the least number of rounds needed so that every vertex is covered by a fire source in some round. In round i a new source is placed on a vertex, and after being placed a fire spreads one step per round. Equivalently, writing $B(v, r)$ for the closed ball of radius r around v ,

$$b(G) = \min \left\{ k \mid \exists v_1, \dots, v_k \in V(G) : \bigcup_{i=1}^k B(v_i, k-i) = V(G) \right\}. \quad (1)$$

Thus for k rounds the radii are exactly $k-1, k-2, \dots, 0$, assigned in that order to the chosen centres.

The conjecture under consideration asserts:

The burning number of the $n \times n$ square grid $P_n \square P_n$ is exactly $\lceil 2\sqrt{n} - 1 \rceil$, and for a general $m \times n$ grid with $m \leq n$ it is given by an explicit piecewise correction of this formula.

The word “exactly” is essential: the claim is an identity for every n , not an asymptotic statement. We show that the identity already fails at $n = 4$.

2 The claimed value at $n = 4$ is wrong

For $n = 4$ the formula gives

$$\lceil 2\sqrt{4} - 1 \rceil = \lceil 4 - 1 \rceil = 3.$$

We prove that no 3-round cover of the 4×4 grid exists, and then exhibit a 4-round cover.

Throughout, identify the 16 cells of the 4×4 grid with pairs $(i, j) \in \{0, 1, 2, 3\}^2$, where i is the row and j the column, and use the Manhattan distance

$$\text{dist}((i, j), (i', j')) = |i - i'| + |j - j'|.$$

For a centre c and radius r , $B(c, r)$ denotes the set of cells at distance at most r from c .

Proposition 2. *There do not exist three cells c_0, c_1, c_2 with*

$$B(c_0, 2) \cup B(c_1, 1) \cup B(c_2, 0) = V(P_4 \square P_4).$$

Proof. By (1) with $k = 3$ the radii are forced to be 2, 1, 0, in that order, so it suffices to rule out every ordered triple (c_0, c_1, c_2) of cells, each lying in $\{0, 1, 2, 3\}^2$, of which there are $16^3 = 4096$. We checked all 4096 triples exhaustively by computer: for each triple we formed the bitmask of $B(c_0, 2) \cup B(c_1, 1) \cup B(c_2, 0)$ and compared it with the full 16-bit mask $2^{16} - 1$; no triple produced the full mask. The computation is purely combinatorial (no search heuristics), so it is a complete proof of the proposition. It is reproduced by `reproduce.py`, and formalised in Lean 4 in `lean4/Main.lean` (`theorem no_three_cover`, proved by exhaustive `decide` over all 4096 cases). $\square$

Remark 3. One can also see the obstruction by hand. A ball of radius 2 covering a corner cell can be assumed, up to the symmetries of the square, to be centred at a corner; the resulting reduced case analysis still leaves a nonempty residue that a radius-1 and a radius-0 ball cannot cover. The exhaustive enumeration above avoids any appeal to such a symmetry reduction.

Proposition 4 (A 4-round cover). *The four balls*

$$B((0,0),3), \quad B((0,2),2), \quad B((3,2),1), \quad B((2,3),0)$$

cover all 16 cells of the 4×4 grid.

Proof. Direct verification. The covered cells, with the index of the round at which each cell is first covered, are

$$\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 4 \\ 1 & 3 & 3 & 3 \end{array}$$

so every one of the 16 cells is covered. For example $B((0,0),3)$ alone covers the ten cells with $\text{dist}((i,j), (0,0)) \leq 3$: the four cells of row 0, the three cells $(1,0), (1,1), (1,2)$ of row 1, the two cells $(2,0), (2,1)$ of row 2, and $(3,0)$. Round 2's ball then covers the remaining cells of the upper part including $(1,3)$ and $(2,2)$, round 3's ball covers $(3,1), (3,2), (3,3)$, and round 4's single source covers $(2,3)$. $\square$

Corollary 5. $b(P_4 \square P_4) = 4$, whereas the conjectured formula gives 3. Hence

$$b(P_4 \square P_4) = 4 \neq 3 = \lceil 2\sqrt{4} - 1 \rceil.$$

In particular the conjecture is false as stated.

3 The failure is not an isolated accident: $n = 6$

For $n = 6$ the formula gives

$$\lceil 2\sqrt{6} - 1 \rceil = \lceil 4.8989 \dots - 1 \rceil = \lceil 3.8989 \dots \rceil = 4.$$

A 4-round cover would require four cells c_0, c_1, c_2, c_3 with

$$B(c_0, 3) \cup B(c_1, 2) \cup B(c_2, 1) \cup B(c_3, 0) = V(P_6 \square P_6).$$

An exhaustive check of all $36^4 = 1,679,616$ ordered quadruples of centres shows that no such cover exists. Therefore

$$b(P_6 \square P_6) \geq 5 > 4 = \lceil 2\sqrt{6} - 1 \rceil,$$

so the closed form fails again. This is reproduced by `reproduce.py` (running time well under a second).

4 Where the formula does hold

To be fair to the conjecture, the closed form is correct for the small values $n = 1, 2, 3, 5$. An exhaustive computation of $b(P_n \sqcap P_n)$ for $n \leq 5$ gives:

n	1	2	3	4	5
$b(P_n \sqcap P_n)$	1	2	3	4	4
$\lceil 2\sqrt{n} - 1 \rceil$	1	2	3	3	4
match?	✓	✓	✓	×	✓

Thus the formula happens to hold at $n = 1, 2, 3, 5$ but fails at $n = 4$ and $n = 6$. The statement of the conjecture uses the phrase “is exactly”, so it asserts an identity for all n ; a large- n or asymptotic reading is not what the text says, and would in any case not rescue the claim at $n = 4$ and $n = 6$. The conjecture is therefore **refuted**.

5 Formal and computational verification

- `lean4/Main.lean` formalises the refutation in core Lean 4 (only `import Std`, no `Mathlib`, no `sorry`, no `axiom`, no `native_decide`). It represents the 4×4 grid as `Fin 4 × Fin 4`, encodes a set of cells as a 16-bit `Nat` bitmask, and defines a Boolean predicate `covers` on a list of centres and a list of radii. The theorem `no_three_cover` states that the number of ordered triples (c_0, c_1, c_2) whose balls of radii 2, 1, 0 cover all 16 cells is 0, and is proved by exhaustive `decide` over the 4096 cases. `four_cover_exists` gives the explicit witness above, and `conjecture_00000003949_false` collects the formula value 3, the non-existence of a 3-round cover, the existence of a 4-round cover, and $3 \neq 4$. `lean4/Check.lean` audits the axioms of all three theorems.
- `reproduce.py` (standard library only) brute-forces $b(P_n \sqcap P_n)$ for $n = 1, \dots, 5$, checks that no 4-round cover of the 6×6 grid exists, prints the comparison with $\lceil 2\sqrt{n} - 1 \rceil$, and prints the explicit 4-source witness together with the covered cells.